

Capacity of Fading Broadcast Channels with Quality-of-Service Constraints

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Abstract. For wireless fading broadcast channels, different capacity configurations such as ergodic, outage, minimum rate, or limited-jitter can be represented by a triplet of quality-of-service parameters from each user: maximum rate, minimum rate, and shortage probability. For each fading state, the channel capacity is obtained from evaluating a finite set of possible extreme points. Optimal power allocation strategy across fading states is shown to be water-filling with rates determined by the effective noise of the corresponding fading states, then combined with constant rate power allocation. Shortage capacity can be similarly obtained by removing the minimum rate constraints when the system is in one of the shortage fading states.

Index Terms: Broadcast channels, capacity region, quality-of-service constraints, fading channels.

I. Introduction

SUPPORTING real-time, multiuser, and heterogeneous traffic such as combinations of multimedia, data, and control signals over wireless communication channels has been an active research topic. In multiuser wireless communications, a broadcast channel refers to a transmitter sending independent information to multiple users simultaneously, as in the downlink communication from a base station to a group of mobile terminal users. The received signal strength fluctuates randomly over time due to scattering, and the constructive and destructive interference of the transmitted signal. This effect of time-varying received signal strength is called fading. For instance, in a favorable fading state when the received signal strength is high, data can be sent at a high rate using little transmit power. On the contrary, in a fading state when signal strength is low, data transmission rate suffers, or it requires large transmit power to boost the transmission rate to an acceptable level. Different applications have different tolerance against such random channel fluctuations. Hence in addition to an average transmission rate, quality-of-service (QoS) parameters such as delay bound, minimum throughput, maximum jitters or outage percentage can be used to fully specify the communications requirements of an application. For example, file downloads might require high average transmission rates, but otherwise have no delay constraints. On the other hand, voice-streaming applications need only moderate average rates, yet impose stringent delay constraint requirements.

The capacity of a channel determines the maximum rate data can be sent to the receiver with arbitrarily small probability of error. Ergodic capacity is the long term average transmission rate over all fading states. Outage capacity is likewise defined, with the additional requirement that the transmitter has to maintain a constant rate, except for a given fraction of time when transmission is allowed to be suspended (typically during unfavorable fading states). As a special case, zero-outage capacity allows no transmission suspension, and thus requires a constant rate at all times. Finally, minimum rate capacity is the average rate achieved with the constraint that a given minimum rate for each user has

to be maintained over all fading states.

While capacity for fading broadcast channels with different constraints are known: ergodic capacity [1], [2], outage capacity [3], and minimum rate capacity [4], this analysis focuses on the scenario where each user is able to specify a set of parameters that represents the QoS requirements on the channel. The user specifies a maximum rate needed by the applications, a minimum rate required, and a shortage probability over which the minimum rate stipulation might be waived. Effectively, when in shortage, the system resorts to a best-service resource provisioning scheme, in which minimum rates are no longer guaranteed. Similar to [3], there is also a system-wide parameter to specify whether common shortage mode, where all users declare shortage simultaneously, or independent shortage mode, where users declare shortage separately, is used.

This model is motivated by the desire to support heterogeneous applications with different QoS requirements over a common wireless channel. For example, delay-insensitive applications such as batch file transfers seek to maximize ergodic capacity with little QoS constraints. On the other hand, an adaptive audio/video streaming application would operate with a minimum rate needed for legible communication up to a maximum rate in which the source content is encoded. Any rate above this maximum does not provide additional utility. Whereas for wireless control applications, a fixed rate (therefore fixed delay) is much more desirable than raw capacity, hence the user would wish to set the minimum rate the same as the maximum rate to eliminate jitters.

The QoS requirement parameters are explained in Section II. Next, Section III presents the discrete-time memoryless broadcast channel system model used in the analysis. Section IV considers the scenario where the broadcast channel is static (i.e., no fading), while Section V extends the analysis results for a fading channel. Effects of shortage on capacity and power allocation is investigated in Section VI. Numerical results are presented in Section VII, and conclusions follow in Section VIII.

II. Quality-of-Service Constraints

Each user will specify the QoS requirements in a triplet of maximum rate R^m , minimum rate R^n , and shortage probability q .

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Let (R_j^m, R_j^n, q_j) be the QoS parameters given by user j , with $0 \leq R_j^n \leq R_j^m$ and $0 \leq q_j \leq 1$. Specifying a maximum rate means user j derives no additional utility beyond the rate R_j^m . The maximum rate may represent system bottlenecks (e.g., limited encoder/decoder processing rate, buffer access speed) such that even if additional power is available, the system still cannot transmit to the user at a higher rate.

On the other hand, setting a minimum rate with shortage probability means user j must receive rate equal to or above R_j^n with probability $1 - q_j$. With the remainder probability q_j , the system can declare shortage and the minimum rate is no longer guaranteed. Shortage capacity differs from outage capacity [3] in that when system is in shortage, while it is short of providing a minimum rate guarantee, transmission might still take place at a lower rate. Whereas when system is in outage, transmission is suspended completely. If transmission power is set to zero when system is in shortage, then shortage capacity reduces to outage capacity.

Besides the QoS requirements triplet (R_j^m, R_j^n, q_j) from each user, a system-wide parameter shortage mode is also to be specified. In common shortage mode, all users are required to declare shortage simultaneously, whereas independent shortage mode allows each user to have different shortage fading states. Naturally, if common shortage is specified, it is assumed that q_j 's of all users are the same: $q_1 = \dots = q_M = q$.

The QoS parameters can be set to represent a variety of channel capacity configurations. For example, having a minimum rate of zero and maximum rate of infinity leads to the ergodic capacity, while setting minimum rate the same as maximum rate represents zero-outage capacity. Different capacity configurations represented by the QoS parameters are shown in Table I.

Capacity Configuration	QoS Parameters
Ergodic [1], [2]	$(R^m, R^n, q) = (\infty, 0, 0)$
Zero-outage [3]	$R^m = R^n, q = 0$
Minimum rate [4]	$(R^m, R^n, q) = (\infty, r_{min}, 0)$
Outage [3]	$R^m = R^n, q = p_{out}$
Limited-jitter	$R^m - R^n = r_{jitter}$

TABLE I

CHANNEL CAPACITY CONFIGURATIONS REPRESENTED BY QOS PARAMETERS.

At first, it might not be intuitive how having a maximum rate parameter would represent a constraint to the system. Since a capacity region encompasses all achievable rates, it would appear that the maximum rate requirements would simply crop the portion of the capacity region that is beyond the respective maximum rates. However, since the maximum rate constraint limits the transmission rate in *every* fading state (instead of limiting the *average* transmission rate), it effectively restricts the transmitter from taking advantage of favorable fading states to transmit at high data rates when noise levels are low.

For example, Fig. 1 shows the ergodic, zero-outage, and minimum rate capacity regions of a two-user fading broadcast channel. The minimum rate requirements are R_1^n and R_2^n , for user 1 and user 2, respectively. As described in [4], the minimum rates have to lie within the zero-outage capacity region; otherwise, the requirements are not achievable. Compared to the ergodic

capacity, imposing minimum rate constraints can only reduce the capacity region. In Fig. 2, it shows the QoS constraint capacity region for the same fading broadcast channel. The minimum rate constraints R_1^n and R_2^n are the same, but in this case there is also a maximum rate constraint R_1^m for user 1. The maximum rate constraint R_1^m lies completely outside the ergodic region. However, since capacity is the average rate over all fading states, without the constraint it is possible for the transmitter to send to user 1 at rate higher than R_1^m in some fading states. The maximum rate constraint imposes a limit on the transmission rate under those scenarios, and therefore has the effect of further reducing the capacity region.

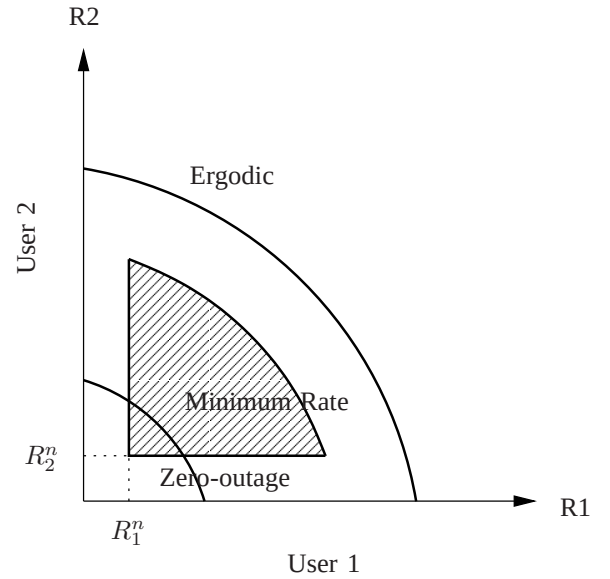


Fig. 1. Ergodic, zero-outage, and minimum rate capacity regions.

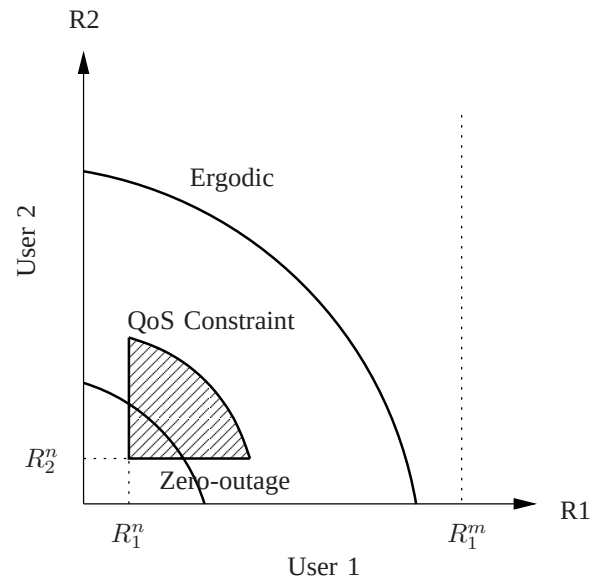


Fig. 2. Ergodic, zero-outage, and QoS constraint capacity regions.

III. System Model

The analysis adopts the same system model as defined in [2]–[4]. It assumes a discrete-time flat-fading Gaussian broadcast channel, with perfect channel state information at the transmitter and at all receivers. The transmitter varies its transmit power (thus controlling the transmission rate) under an average power constraint, and superposition coding with successive decoding [5] is used.

At each time instance i , the transmitter sends a signal $X_j(i)$ with power $p_j(i)$ to user j over bandwidth B . The average total transmission power over time cannot exceed $\bar{P}$. To characterize the channel, let v_j be the noise density of user j 's channel, and $\sqrt{g_j(i)}$ its time-varying channel gain at time i . Then combine the channel gain and noise density to form the time-varying noise density $n_j(i) = v_j/g_j(i)$; since $n_j(i)$ incorporates the time-varying channel state, it will also be referred to as a fading state. Signals to all M users are sent simultaneously, hence the received signal for user j becomes

$$Y_j(i) = \sum_{k=1}^M X_k(i) + z_j(i) \quad (1)$$

where $z_j(i)$ is a Gaussian random variable with zero mean and variance $n_j(i)B$. It is assumed that the noise density vector $\mathbf{n}(i) = (n_1(i), \dots, n_M(i))$ are known to the transmitter and all receivers at each time instance i .

Using superposition coding with successive decoding, the rate for user j at fading state $\mathbf{n}$ is

$$R_j(\mathbf{n}) = B \log \left(1 + \frac{p_j(\mathbf{n})}{n_j B + \sum_{i=1}^M p_i(\mathbf{n}) \mathbf{1}[n_j > n_i]} \right) \quad (2)$$

where $\mathbf{1}[\cdot]$ is the indicator function. Since B is simply a constant scaling factor, it can be assumed to be 1 without loss of generality. This analysis will focus on the case where the channel has two users ($M = 2$).

IV. Additive White Gaussian Noise Channel

First consider an additive white Gaussian noise (AWGN) channel where there is no fading. In this case, n_1, n_2 are constants with respect to time. Furthermore, assume $n_2 > n_1$ (user 2 has noisier channel); in the case of $n_1 > n_2$, the derivation remains valid with the subscripts 1, 2 reversed. Let user 1 and user 2 has QoS requirements $(R_1^m, R_1^n, 0)$ and $(R_2^m, R_2^n, 0)$, respectively (since channel has no fading, set shortage probability q_1 and q_2 to 0). To determine channel capacity, it is needed to maximize the achievable rates subject to the total power and QoS constraints:

$$\begin{aligned} \max_{p_1, p_2} \quad & \mu_1 \log \left(1 + \frac{p_1}{n_1} \right) + \mu_2 \log \left(1 + \frac{p_2}{p_1 + n_2} \right) \\ \text{subject to: } & p_1 + p_2 \leq P, (R_1^m, R_1^n, 0), (R_2^m, R_2^n, 0) \end{aligned} \quad (3)$$

where p_1 is the power allocated to user 1, and p_2 the power allocated to user 2. The proportion factors μ_1 and μ_2 allow achievable rates of user 1 and user 2 to be weighted differently; $0 < \mu_1 < 1$, and $\mu_2 = 1 - \mu_1$. For instance, $\mu_1 = \mu_2 = 1/2$ corresponds to the case when the metric of interest is the sum-rate $R_1 + R_2$, and both users' rates are weighted equally. Note that from the definition $\mu_1 \neq 0$ and $\mu_2 \neq 0$: This is to simplify

the boundary conditions in the maximization; as each factor can be arbitrarily close to zero, no generality is lost.

The channel capacity is a function of (p_1, p_2) , the noise density vector $\mathbf{n} = (n_1, n_2)$, and the total available power P . In an AWGN channel, $\mathbf{n}$ is known and remains constant. Suppose total power P is available for transmission, then the problem is to find out how to optimally divide up P between the two users; i.e., what is p_1^* and p_2^* ? Let the function $F_A(P, \mathbf{n})$ denote this optimal power allocation between the two users. Given P and $\mathbf{n}$, $F_A(\cdot)$ returns the optimal p_1^* and p_2^* , which can then be substituted back in (3) to obtain the capacity achieved under optimal power allocation. Therefore, the optimal power allocation function is

$$(p_1^*, p_2^*) = F_A(P, \mathbf{n}), \quad (4)$$

and the respective achieved capacity is

$$R_C(P, \mathbf{n}) = \max_{p_1, p_2} R((p_1, p_2), P, \mathbf{n}) \quad (5)$$

$$= R(F_A(P, \mathbf{n}), P, \mathbf{n}). \quad (6)$$

Note that $R((p_1, p_2), P, \mathbf{n})$ represents an achievable rate with *some* power allocation p_1 and p_2 . When the power allocation is optimal, capacity is therefore achieved and is denoted by $R_C(\cdot)$. Since power allocation is already determined, $R_C(\cdot)$ is only a function of P and $\mathbf{n}$.

Successive decoding allows user 1 to subtract out the signal power of user 2. Therefore, user 1 is not affected by user 2. To user 2, however, user 1's signal power p_1 is effectively noise. For user 2 to maintain a constant rate R_2^m , as derived in [4], its power p_2 needs to be accordingly increased whenever p_1 is increased:

$$p_2 = (e^{R_2^m} - 1)p_1 + n_2(e^{R_2^m} - 1). \quad (7)$$

Likewise, if a constant rate of R_2^n is to be maintained:

$$p_2 = (e^{R_2^n} - 1)p_1 + n_2(e^{R_2^n} - 1). \quad (8)$$

For convenience, define the following constants:

$$\begin{aligned} P_1^m &\triangleq n_1(e^{R_1^m} - 1) \\ P_1^n &\triangleq n_1(e^{R_1^n} - 1) \\ G_m &\triangleq e^{R_2^m} - 1 \\ G_n &\triangleq e^{R_2^n} - 1 \\ P_2^m &\triangleq G_m(P_1^m + n_2) \\ P_2^n &\triangleq G_n(P_1^n + n_2). \end{aligned}$$

Then the constraints of (3) can be expressed in terms of the following five inequalities:

$$p_1 + p_2 \leq P \quad (9)$$

$$p_1 \leq P_1^m \quad (10)$$

$$p_1 \geq P_1^n \quad (11)$$

$$p_2 \leq G_m(p_1 + n_2) \quad (12)$$

$$p_2 \geq G_n(p_1 + n_2) \quad (13)$$

Note that all of the constraints are linear in p_1, p_2 . Let Ω denote the region over p_1, p_2 where all constraints are met. Since $P \geq 0$ and $0 \leq R_j^n \leq R_j^m$ for each user $j = 1, 2$, it implies $P_1^m \geq P_1^n$, $G_m \geq G_n$ and $P_2^m \geq P_2^n$. These conditions lead to a contiguous and closed region Ω on the p_1 - p_2 plane, as shown in Fig. 3. Let

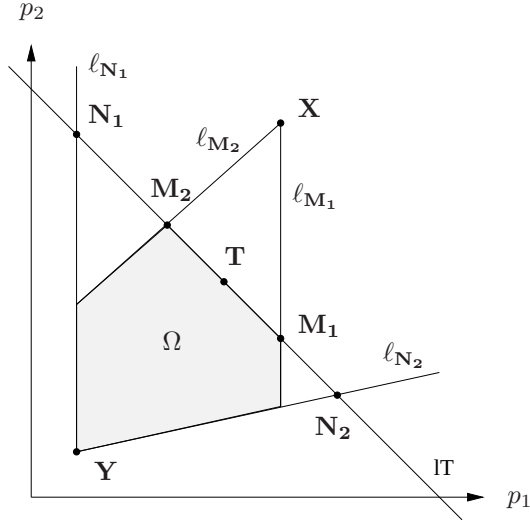


Fig. 3. QoS constraints on power allocation for user 1, 2.

the lines ℓ_T , ℓ_{M_1} , ℓ_{N_1} , ℓ_{M_2} and ℓ_{N_2} denote the boundaries of Ω , which correspond, respectively, to constraints (9)–(13) with equality being taken, as summarized in Table II below.

Line	Constraint	Constraint Type
ℓ_T	(9)	total average power
ℓ_{M_1}	(10)	maximum rate for user 1
ℓ_{N_1}	(11)	minimum rate for user 1
ℓ_{M_2}	(12)	maximum rate for user 2
ℓ_{N_2}	(13)	minimum rate for user 2

TABLE II
CONSTRAINT BOUNDARIES.

An extreme point is defined by the intersection of two boundaries. Note that a point (p_1, p_2) can be represented as a vector $\mathbf{p} = p_1 \mathbf{i} + p_2 \mathbf{j}$, where $\mathbf{i}$, $\mathbf{j}$ are unit vectors in the p_1 , p_2 directions, respectively. Six extreme points $\mathbf{M}_1$, $\mathbf{N}_1$, $\mathbf{M}_2$, $\mathbf{N}_2$, $\mathbf{X}$ and $\mathbf{Y}$ are of particular interest; their definitions are tabulated in Table III.

Point	Intersection	Constraint Type
$\mathbf{M}_1$	ℓ_T, ℓ_{M_1}	maximum rate for user 1
$\mathbf{N}_1$	ℓ_T, ℓ_{N_1}	minimum rate for user 1
$\mathbf{M}_2$	ℓ_T, ℓ_{M_2}	maximum rate for user 2
$\mathbf{N}_2$	ℓ_T, ℓ_{N_2}	minimum rate for user 2
$\mathbf{X}$	ℓ_{M_1}, ℓ_{M_2}	minimum rate for both users
$\mathbf{Y}$	ℓ_{N_1}, ℓ_{N_2}	minimum rate for both users

TABLE III
CONSTRAINT BOUNDARY EXTREME POINTS.

The channel achievable rate function

$$R(\mathbf{p}) = \mu_1 \log \left(1 + \frac{p_1}{n_1} \right) + \mu_2 \log \left(1 + \frac{p_2}{p_1 + n_2} \right), \quad (14)$$

being differentiable everywhere, is differentiable on the closed region Ω . Therefore, $R(\mathbf{p})$ must take on an absolute maximum on Ω , which can be found as follows:

- 1) First calculate the the gradient of $R(\mathbf{p})$:

$$\begin{aligned} \nabla R(\mathbf{p}) &= \frac{\partial R}{\partial p_1}(\mathbf{p}) \mathbf{i} + \frac{\partial R}{\partial p_2}(\mathbf{p}) \mathbf{j} \\ &= \left(\frac{\mu_1}{p_1 + n_1} - \frac{\mu_2 p_2}{(p_1 + n_2)(p_1 + p_2 + n_2)} \right) \mathbf{i} \\ &\quad + \frac{\mu_2}{p_1 + p_2 + n_2} \mathbf{j}. \end{aligned} \quad (15)$$

The gradient, defined everywhere, is never $\mathbf{0}$. In particular, since $\mu_2 > 0$, its $\mathbf{j}$ -component will always be positive. Therefore, the maximum point of $R(\mathbf{p})$ is on one of the boundaries of Ω .

- 2) Consider the boundaries ℓ_{M_2} and ℓ_{N_2} . Define unit vectors $\mathbf{u}_m$, $\mathbf{u}_n$, respectively, to have the same directions as ℓ_{M_2} , ℓ_{N_2} (in the increasing p_1 - p_2 polarity):

$$\mathbf{u}_m = \frac{\mathbf{i} + G_m \mathbf{j}}{\sqrt{(G_m)^2 + 1}}, \quad (17)$$

$$\mathbf{u}_n = \frac{\mathbf{i} + G_n \mathbf{j}}{\sqrt{(G_n)^2 + 1}}. \quad (18)$$

The directional derivatives of $R(\mathbf{p})$ with respect to $\mathbf{u}_m$ and $\mathbf{u}_n$, respectively, are given by the dot products:

$$R'_{\mathbf{u}_m}(\mathbf{p}) = \nabla R(\mathbf{p}) \cdot \mathbf{u}_m \quad (19)$$

$$= \frac{1}{\sqrt{(G_m)^2 + 1}} \frac{\mu_1}{p_1 + n_2}, \quad (20)$$

$$R'_{\mathbf{u}_n}(\mathbf{p}) = \nabla R(\mathbf{p}) \cdot \mathbf{u}_n \quad (21)$$

$$= \frac{1}{\sqrt{(G_n)^2 + 1}} \frac{\mu_1}{p_1 + n_2}. \quad (22)$$

Since both $R'_{\mathbf{u}_m}(\mathbf{p})$ and $R'_{\mathbf{u}_n}(\mathbf{p})$ are always positive, $R(\mathbf{p})$ is monotonically increasing in the directions of $\mathbf{u}_m$ and $\mathbf{u}_n$. Therefore, if the maximum point lies on ℓ_{M_2} or ℓ_{N_2} , it has to be one of the extreme points $\mathbf{M}_2$, $\mathbf{N}_2$, or $\mathbf{X}$.

- 3) Next consider the boundaries ℓ_{M_1} and ℓ_{N_1} . Both of them are parallel to the p_2 axis, hence the derivative of $R(\mathbf{p})$ in the direction of $\mathbf{j}$ is

$$R'_j(\mathbf{p}) = \nabla R(\mathbf{p}) \cdot \mathbf{j} \quad (23)$$

$$= \frac{\mu_2}{p_1 + p_2 + n_2}. \quad (24)$$

Similarly, since the directional derivative $R'_j(\mathbf{p})$ is always positive, $R(\mathbf{p})$ is monotonically increasing in the direction of $\mathbf{j}$. Therefore, if the maximum point lies on ℓ_{M_1} or ℓ_{N_1} , it has to be one of the extreme points $\mathbf{M}_1$ or $\mathbf{N}_1$.

- 4) If the maximum point does not lie on ℓ_{M_2} , ℓ_{N_2} , ℓ_{M_1} , or ℓ_{N_1} , then it has to be on the boundary ℓ_T within the open interval between points $\mathbf{M}_1$ and $\mathbf{M}_2$. Denote such point as $\mathbf{T}$. Then optimization (3) reduces to a standard one-dimensional maximization problem. The derivative of the channel capacity along ℓ_T with respect to p_1 is

$$\frac{dR_{\ell_T}}{dp_1}(p_1) = \frac{(\mu_1 - \mu_2)p_1 + \mu_1 n_2 - \mu_2 n_1}{(p_1 + n_1)(p_1 + n_2)}. \quad (25)$$

Since $n_2 > n_1$, the derivative is always positive for $\mu_1 \geq \mu_2$. In such case $R_{\ell_T}(p_1)$ is monotonically increasing in the

direction of p_1 , so $\mathbf{M}_1$ or $\mathbf{N}_2$ are the possible maximum points rather than $\mathbf{T}$. For $\mu_1 < \mu_2$, $\mathbf{T}$ is given by

$$\mathbf{T} = \left(\frac{\mu_1 n_2 - \mu_2 n_1}{\mu_2 - \mu_1}, P - \frac{\mu_1 n_2 - \mu_2 n_1}{\mu_2 - \mu_1} \right). \quad (26)$$

- 5) Finally, when available total power equals the minimum power requirement $P = P_1^n + P_2^n$. All of the above operating points close in to meet at the minimum rate extreme point $\mathbf{Y}$.

Therefore, there are only seven potential maximum point candidates: $\mathbf{M}_1, \mathbf{N}_1, \mathbf{M}_2, \mathbf{N}_2, \mathbf{T}, \mathbf{X}$ and $\mathbf{Y}$. Thus the maximum of $R(\mathbf{p})$ can be obtained by evaluating the candidate points that lie on the boundaries of Ω , and choose the one that yields the largest value, i.e.,

$$F_A(P, \mathbf{n}) = \mathbf{v} \text{ such that } R(\mathbf{v}) \geq R(\mathbf{v}') \forall \mathbf{v}' \neq \mathbf{v}, \quad (27)$$

where $\mathbf{v}, \mathbf{v}' \in \{\mathbf{M}_1, \mathbf{N}_1, \mathbf{M}_2, \mathbf{N}_2, \mathbf{T}, \mathbf{X}, \mathbf{Y}\} \cap \Omega$.

The coordinates of the possible maximum points are presented in Table IV. To evaluate the channel capacity at the possible maximum points, substitute the coordinates into (14) to obtain:

$$R(\mathbf{M}_1) = \mu_2 \log \left(1 + \frac{P - P_1^m}{P_1^m + n_2} \right) + \mu_1 R_1^m \quad (28)$$

$$R(\mathbf{N}_1) = \mu_2 \log \left(1 + \frac{P - P_1^n}{P_1^n + n_2} \right) + \mu_1 R_1^n \quad (29)$$

$$R(\mathbf{M}_2) = \mu_1 \log \left(1 + \frac{P - G_m n_2}{(G_m + 1)n_1} \right) + \mu_2 R_2^m \quad (30)$$

$$R(\mathbf{N}_2) = \mu_1 \log \left(1 + \frac{P - G_n n_2}{(G_n + 1)n_1} \right) + \mu_2 R_2^n \quad (31)$$

$$R(\mathbf{T}) = \mu_2 \log \left(1 + \frac{P - \frac{\mu_2 n_1 - \mu_1 n_2}{\mu_1 - \mu_2}}{\frac{\mu_2}{\mu_1 - \mu_2}(n_1 - n_2)} \right) + \mu_1 \log \left(1 + \frac{\mu_2 n_1 - \mu_1 n_2}{(\mu_1 - \mu_2)n_1} \right) \quad (32)$$

$$R(\mathbf{X}) = \mu_1 R_1^m + \mu_2 R_2^m \quad (33)$$

$$R(\mathbf{Y}) = \mu_1 R_1^n + \mu_2 R_2^n \quad (34)$$

Then the channel capacity associated with the optimal power allocation is

$$R_C(P, \mathbf{n}) = R(F_A(P, \mathbf{n}), P, \mathbf{n}), \quad (35)$$

which is given by one of the equations in (28)–(34).

V. Fading Channel

In this section, zero-shortage capacity of a fading channel will be studied. Effects of shortage are analyzed in Section VI. For a given total power $P(\mathbf{n})$ and noise density $\mathbf{n}$, optimally allocating the power between the users is solved in Section IV; the power allocation between user 1 and user 2 is given by (27) and the capacity achieved is the maximum of (28)–(34). Hence, the remaining step is to determine the optimal power allocation across fading states $\mathbf{n}$.

In a fading channel, n_1, n_2 are random variables with known joint probability distribution. Consider based on the noise density vector $\mathbf{n} = (n_1, n_2)$, the total power $P(\mathbf{n})$ can be varied. Partition the noise density vector into states $\{S_{M_1}, S_{N_1}, S_{M_2}, S_{N_2}, S_T, S_X, S_Y\}$, where S_v corresponds to the values of $\mathbf{n}$ that results

in the channel capacity taking on maximum at $\mathbf{v}$ for a given total power P , i.e.,

$$\mathbf{n} \in S_v \text{ if } F_A(P, \mathbf{n}) = \mathbf{v}, \quad (36)$$

where $\mathbf{v} \in \{\mathbf{M}_1, \mathbf{N}_1, \mathbf{M}_2, \mathbf{N}_2, \mathbf{T}, \mathbf{X}, \mathbf{Y}\}$.

Therefore, the channel capacity for a given noise density state S_v and total power P is given by (28)–(34). For states S_X and S_Y , the channel has constant rates. For the other states, the channel capacity has the following form:

$$R_C(P(\mathbf{n}), \mathbf{n}) = A_v \log \left(1 + \frac{P(\mathbf{n}) - B_v}{C_v} \right) + D_v \quad (37)$$

where A_v, B_v, C_v and D_v are parameters specific to the noise density state S_v . The values of these parameters are tabulated in Table V.

To find the optimal power allocation strategy for each $\mathbf{n}$, i.e., to determine $P(\mathbf{n})$, it is needed to maximize the following:

$$\max_{P(\mathbf{n})} E_{\mathbf{n}} [R(p_1(\mathbf{n}), p_2(\mathbf{n})), P(\mathbf{n}), \mathbf{n}] \quad (38)$$

$$= \max_{P(\mathbf{n})} E_{\mathbf{n}} \left[\mu_1 \log \left(1 + \frac{p_1(\mathbf{n})}{n_1} \right) + \mu_2 \log \left(1 + \frac{p_2(\mathbf{n})}{p_1(\mathbf{n}) + n_2} \right) \right] \quad (39)$$

subject to: $E_{\mathbf{n}}[P(\mathbf{n})] \leq \bar{P}$,

where $p_1(\mathbf{n}) + p_2(\mathbf{n}) \leq P(\mathbf{n})$.

For S_X and S_Y , power allocation is dictated by the maximum and minimum rates:

$$P_X(\mathbf{n}) = P_1^m + P_2^m \quad (40)$$

$$P_Y(\mathbf{n}) = P_1^n + P_2^n \quad (41)$$

where $P_X(\mathbf{n}), P_Y(\mathbf{n})$ are the power allocation strategy for states S_X, S_Y , respectively. For the other states, substitute (37) in (38) to obtain:

$$\max_{P(\mathbf{n})} E_{\mathbf{n}} \left[A_v \log \left(1 + \frac{P(\mathbf{n}) - B_v}{C_v} \right) + D_v \right] \quad (42)$$

subject to: $E_{\mathbf{n}}[P(\mathbf{n})] \leq \bar{P}$.

Using Lagrangian method, form

$$J(P(\mathbf{n})) = A_v \log \left(1 + \frac{P(\mathbf{n}) - B_v}{C_v} \right) + D_v - \lambda (E_{\mathbf{n}}[P(\mathbf{n})] - \bar{P}), \quad (43)$$

then set $\frac{\partial J}{\partial P(\mathbf{n})}$ to zero to obtain:

$$P(\mathbf{n}) = \begin{cases} A_v \frac{1}{\lambda} - (C_v - B_v) & A_v \frac{1}{\lambda} \geq C_v - B_v \\ 0 & \text{else} \end{cases} \quad (44)$$

where $\frac{1}{\lambda}$ is the water-filling level. Define effective noise $n_v^*(n_1, n_2) = C_v - B_v$. The values of A_v and n_v^* for each state are summarized in Table VI. Notice that states S_{M_1}, S_{N_2} and S_T all have $A_v = \mu_2$ and the same effective noise $n_v^* = n_2$. Therefore, there are only three water-filling equations:

$$P_{M_1, N_1, T}(\mathbf{n}) = \left[\mu_2 \frac{1}{\lambda} - n_2 \right]_+ \quad (45)$$

$$P_{M_2}(\mathbf{n}) = \left[\mu_1 \frac{1}{\lambda} - (G_m + 1)n_1 + G_m n_2 \right]_+ \quad (46)$$

$$P_{N_2}(\mathbf{n}) = \left[\mu_1 \frac{1}{\lambda} - (G_n + 1)n_1 + G_n n_2 \right]_+ \quad (47)$$

Point	p_1	p_2	Constraint Type
M_1	P_1^m	$P - P_1^m$	maximum rate for user 1
N_1	P_1^n	$P - P_1^n$	minimum rate for user 1
M_2	$\frac{1}{G_m+1}(P - G_m n_2)$	$\frac{G_m}{G_m+1}(P + n_2)$	maximum rate for user 2
N_2	$\frac{1}{G_n+1}(P - G_n n_2)$	$\frac{G_n}{G_n+1}(P + n_2)$	minimum rate for user 2
T	$\frac{\mu_1 n_2 - \mu_2 n_1}{\mu_2 - \mu_1}$	$P - \frac{\mu_2 n_1 - \mu_1 n_2}{\mu_1 - \mu_2}$	total average power
X	P_1^m	P_2^m	maximum rate for both users
Y	P_1^n	P_2^n	minimum rate for both users

TABLE IV
POSSIBLE MAXIMUM POINTS.

State	A_v	B_v	C_v	D_v
S_{M_1}	μ_2	P_1^m	$P_1^m + n_2$	$\mu_1 R_1^m$
S_{N_1}	μ_2	P_1^n	$P_1^n + n_2$	$\mu_1 R_1^n$
S_{M_2}	μ_1	$G_m n_2$	$(G_m + 1)n_1$	$\mu_2 R_2^m$
S_{N_2}	μ_1	$G_n n_2$	$(G_n + 1)n_1$	$\mu_2 R_2^n$
S_T	μ_2	$\frac{\mu_1 n_2 - \mu_2 n_1}{\mu_1 - \mu_2}$	$\frac{\mu_2}{\mu_1 - \mu_2}(n_1 - n_2)$	$\mu_1 \log \left(1 + \frac{\mu_2 n_1 - \mu_1 n_2}{(\mu_1 - \mu_2)n_1} \right)$

TABLE V
RATE FUNCTION PARAMETERS FOR EACH NOISE DENSITY STATE.

State	A_v	$n_v^*(n_1, n_2)$
S_{M_1}	μ_2	n_2
S_{N_1}	μ_2	n_2
S_{M_2}	μ_1	$(G_m + 1)n_1 - G_m n_2$
S_{N_2}	μ_1	$(G_n + 1)n_1 - G_n n_2$
S_T	μ_2	n_2

TABLE VI

WATER-FILLING PARAMETERS FOR EACH NOISE DENSITY STATE.

Next, it is needed to determine which power allocation equation to use for a given fading state $\mathbf{n}$. Consider the power allocation $P_v(\mathbf{n})$ in the direction n_1 for each state. Effective noise of states S_{M_1} , S_{N_1} , S_T does not depend on n_1 , so allocated power is simply the constant $\mu_2 \frac{1}{\lambda} - n_2$, whereas states S_{M_2} and S_{N_2} have linear power allocation with slopes, respectively, $-(G_m + 1)$ and $-(G_n + 1)$, as illustrated in Fig. 4.

Observe that as given in Table IV, as n_1 increases, T moves from N_2 towards M_2 ; therefore, in the direction of increasing n_1 the states can only appear in this order: $\{S_{N_2}, S_T, S_{M_2}\}$. Coupled with the fact that slope of $P_{M_1, N_1, T}(n_1) \geq$ slope of $P_{N_2}(n_1) \geq$ slope of $P_{M_2}(n_1)$ with respect to n_1 , the water-filling power allocation can be represented compactly by

$$P_W(\mathbf{n}) = \min\{\max\{P_{N_2}(\mathbf{n}), P_{M_1, N_1, T}(\mathbf{n})\}, P_{M_2}(\mathbf{n})\}. \quad (48)$$

Alternatively, in the direction of n_2 , increasing n_2 while keeping n_1 fixed implies the state occurrence order $\{S_{M_2}, S_{N_1}\}$ and $\{S_{M_1}, S_{N_2}\}$. With slope of $P_{N_1}(n_2) >$ slope of $P_{M_2}(n_2)$, and slope of $P_{M_1}(n_2) >$ slope of $P_{N_2}(n_2)$, it leads to the same conclusion. Finally, combine with the maximum and minimum rate constraints, the optimal power allocation strategy is:

$$P^*(\mathbf{n}) = \min\{\max\{P_W(\mathbf{n}), P_Y(\mathbf{n})\}, P_X(\mathbf{n})\}. \quad (49)$$

The water-filling level $\frac{1}{\lambda}$ can be determined by choosing the maximum water-filling level allowed by the average total power

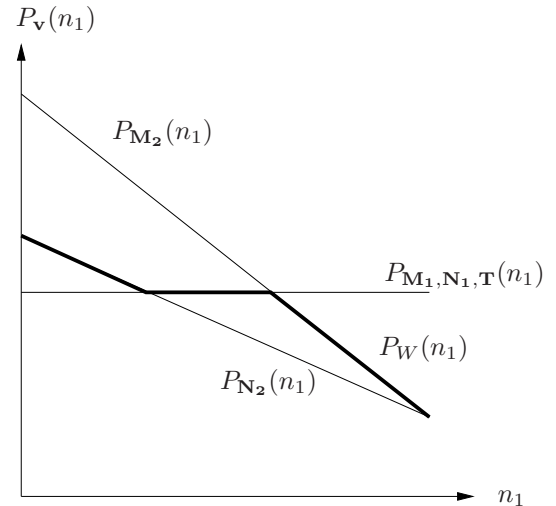


Fig. 4. Power allocation for each noise density state S_v in the direction of n_1 , with the combined water-filling power allocation $P_W(n_1)$ shown in bold.

constraint:

$$\lambda = \lambda^* \text{ such that } \max_{\lambda^*} E_{\mathbf{n}}[P(\mathbf{n}, \lambda^*)] \leq \bar{P}, \quad (50)$$

and the channel capacity achieved is simply the weighted average of the capacity over the fading states:

$$R^* = E_{\mathbf{n}}[R(P^*(\mathbf{n}))]. \quad (51)$$

This is illustrated in Fig. 5, where the dashed line denotes a lower water-filling level than the one in bold.

VI. Shortage Probability

A. Common Shortage

In the common shortage mode, when the system declares shortage for a certain fading state $\mathbf{n}$, the minimum rate constraints

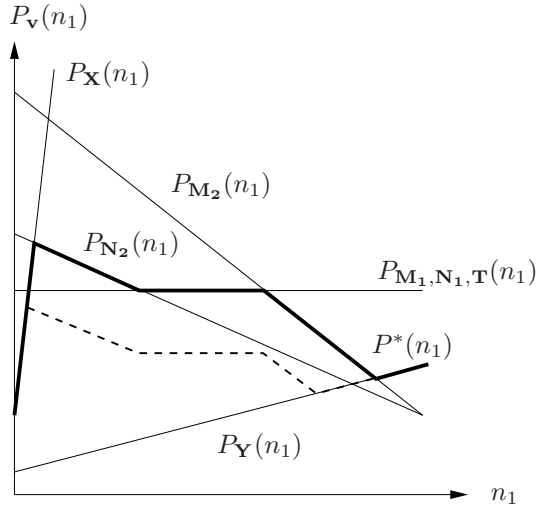


Fig. 5. Water-filling combined with constant rate power allocation. The dashed line denotes a lower water-filling level than the one in bold.

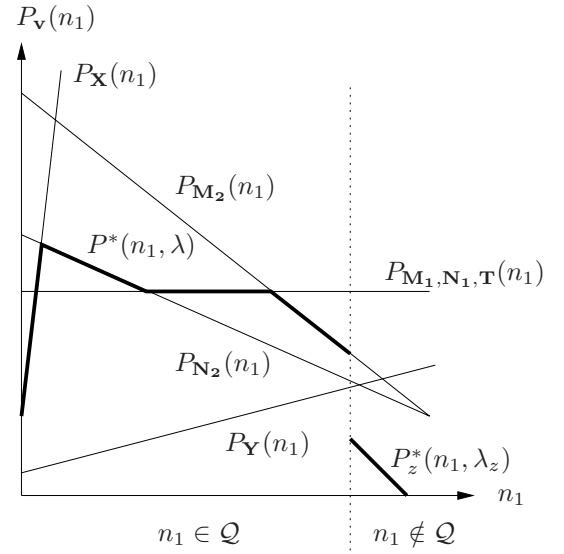


Fig. 6. Power allocation with shortage probability.

for all M users are removed. Naturally, all users have the same shortage probability q . Effectively, this is same optimization problem as before with minimum rates $R_j^n = 0 \forall j = 1, \dots, M$. Therefore, similar procedures from Section IV can be used to deduce the optimal power allocation.

- 1) First calculate the zero-shortage power allocation function as previously derived. Write $P^*(\mathbf{n}, \lambda)$ to denote power allocation in terms of its water-filling level λ , and $P^*(\mathbf{n})$ when λ is determined by the average total power constraint (50).
- 2) Repeat the same calculations with the minimum rate constraints removed, i.e., set $R_j^n = 0 \forall j$; denote such power allocation function obtained as $P_z^*(\mathbf{n}, \lambda_z)$, and when the water-filling level λ_z is determined, as $P_z^*(\mathbf{n})$.
- 3) Select the set of shortage fading states $\mathcal{Q}$ to contain the fading states $\mathbf{n}$ that results in maximum power savings while not exceeding the specified shortage probability q :

$$\max_{\mathcal{Q}} \sum_{\mathbf{n} \in \mathcal{Q}} P^*(\mathbf{n}) - P_z^*(\mathbf{n}) \text{ such that } \sum_{\mathbf{n} \in \mathcal{Q}} \Pr(\mathbf{n}) \leq q \quad (52)$$

- 4) As shown in Fig. 6, the final shortage capacity power allocation strategy $P_{\mathcal{Q}}(\mathbf{n})$ is obtained from combining $P^*(\mathbf{n}, \lambda)$ and $P_z^*(\mathbf{n}, \lambda_z)$ over their respective associated domain of $\mathbf{n}$.

$$P_{\mathcal{Q}}(\mathbf{n}, \lambda, \lambda_z) = \begin{cases} P^*(\mathbf{n}, \lambda) & \mathbf{n} \notin \mathcal{Q}, \\ P_z^*(\mathbf{n}, \lambda_z) & \mathbf{n} \in \mathcal{Q}. \end{cases} \quad (53)$$

The water-filling levels λ, λ_z need to be recalculated to satisfy the average total power constraint (50). However, since the removal of minimum rate constraints represents a discontinuous change of channel capacity with respect to $\mathbf{n}$, the water-filling levels λ and λ_z are independent. After constraint (50) is applied, there still is an extra degree of freedom over division of power between λ and λ_z . Unfortunately, as λ and λ_z have no closed form analytic expressions, standard optimization techniques cannot be applied. Notwithstanding, in practice when the shortage probability q specified is small, λ_z is often zero. This is because small q limits $\mathcal{Q}$ to only contain fading states with large noise density $\mathbf{n}$, over which the optimal power allocation strategy simply is to suspend

transmission. In this case shortage capacity reduces to outage capacity.

B. Independent Shortage

In independent shortage mode, each user specifies a shortage probability q_j , $j = 1, \dots, M$, over which the minimum rate constraint for that user is not guaranteed, i.e., $R_j^n = 0$. Calculating the optimal power allocation for independent shortage capacity is similar to that of common shortage. However, it is needed to compare the zero-shortage power allocation function to that of each of the $2^M - 1$ permutations of different users in shortage. Moreover, it is needed to divide the power among the $2^M - 1$ water-filling levels. As described in Section VI-A, as there is no closed form expression for the channel capacity, standard optimization techniques cannot be readily applied. Further research work is needed to devise the optimal power allocation strategy for this scenario.

VII. Numerical Simulations

In this section simulation results are presented. In all plots it is assumed that the average total transmission power $\bar{P}$ is 10mW, and the channel has a bandwidth B of 100kHz. In Fig. 7, the fading state distribution is: $(n_1 B, n_2 B) = (1 \times 10^{-4} \text{mW}, 1 \text{mW})$ with probability 1/2, and $(n_1 B, n_2 B) = (1 \text{mW}, 1 \times 10^{-4} \text{mW})$ with probability 1/2. The channel has a large 40dB SNR fluctuation between the fading states for each user. As expected, imposing a minimum rate constraint of 300kbps for each of the users reduces the capacity region. If each user also specifies a maximum rate constraint of 1200kbps, capacity region is further reduced. Note that the 1200kbps maximum rate actually lies outside of the ergodic capacity region. However, it still represents a limitation on the system since the transmitter can no longer send at arbitrarily high rates (within transmit power constraint) in a favorable fading state.

In Fig. 8, the channel has the fading distribution: $(n_1 B, n_2 B) = (0.001 \text{mW}, 0.1 \text{mW})$ with probability 1/2, and $(n_1 B, n_2 B)$

= (0.1mW, 0.001mW) with probability 1/2. In this case, the channel has a relatively smaller 20dB fluctuation in SNR between the fading states for each user. Therefore, imposing a minimum rate of 300kbps does not reduce the capacity region drastically. In fact, unlike the previous scenario, specifying a maximum rate constraint of 1200kbps for each user does not reduce the minimum rate capacity. It implies that even without the maximum rate constraint, in every fading state the transmission rate for each user does not exceed 1200kbps, thus having the maximum rate constraint does not represent any additional limitation to the transmitter. When a more restrictive maximum rate of 900kbps is imposed, it can be observed that only then the capacity region is reduced moderately.

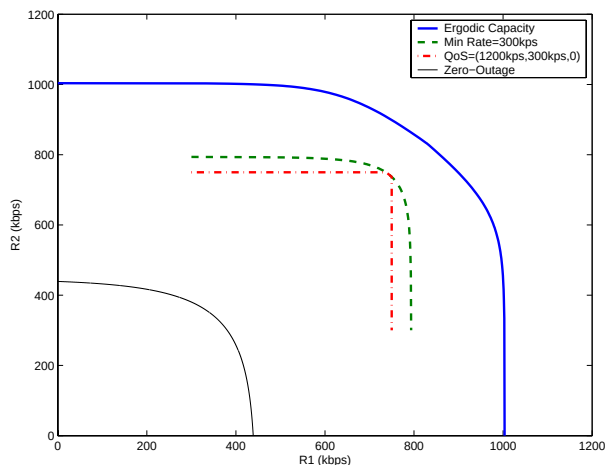


Fig. 7. Capacity regions of a symmetric two-user fading broadcast channel, with 40dB SNR difference between fading states.

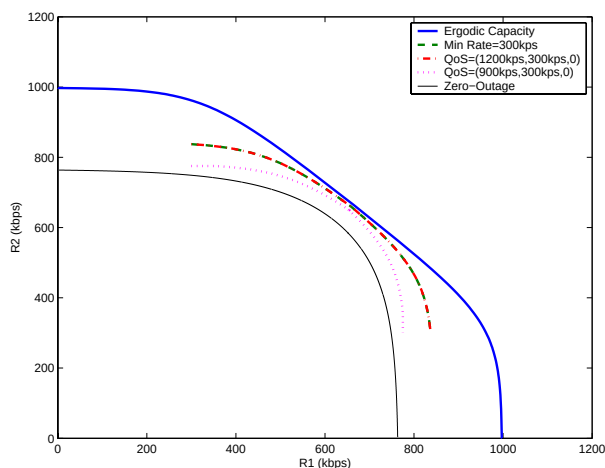


Fig. 8. Capacity regions of a symmetric two-user fading broadcast channel, with 20dB SNR difference between fading states.

VIII. Conclusion

In this analysis it has been shown that a variety of channel capacity configurations, e.g., ergodic, outage, minimum rate, limited-jitter, or a heterogeneous combination of them, can be represented by a triplet of QoS parameters from each user. Optimal power allocation between the users can be readily determined

by evaluating a finite set of possible extreme points. The resulting channel capacity with respect to total power is either constant, or has a common form that is logarithmically proportional to the total power relative to an effective noise. This common capacity expression allows power allocation across fading states to be optimized, which is shown to be water-filling with rates determined by the effective noise of the corresponding fading states, then combined with constant rate power allocation. Finally, it is shown that shortage capacity can be similarly obtained by removing the minimum rate constraints when the system is in one of the shortage fading states. Therefore, the QoS parameter model provides a unifying framework in which channel capacity configurations with heterogeneous requirements from different users can be analyzed to obtain the respective channel capacity and optimal power allocation strategy.

Acknowledgment

The author would like to thank Prof. A. Goldsmith and Nihar Jindal for helpful discussions on the topic.

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